

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Nov 2024

Course Code: INT405R02

Course: MACHINE LEARNING TECHNIQUES

QP No. :U340-7

Duration: 3 hours

Max. Marks:100

PART – A

Answer any FOUR questions

4 x 20 = 80 Marks

1. a) Given the 2D signal represented by the matrix A:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

Compute the Singular Value Decomposition (SVD) to decompose a 2D signal into orthogonal components. (15)

- b) Derive the Gradient Descent optimization algorithm for minimizing the cost function of Univariate Linear Regression models. Provide a detailed mathematical formulation for updating the parameters. (5)

2. a) In SASTRA University, students are enrolled in two types of programs: undergraduate and post graduate. 60 % of the students are enrolled in undergraduate programs and 40 % of the students are enrolled in post graduate programs. Among 10% of undergraduate students receive scholarships and 30 % of post graduate students receive scholarships. Overall, 15 % of all students receive scholarships. A student is selected at random from the university and is found to have received a scholarship.

- i) Calculate the probability that the selected student is enrolled in an undergraduate program. (6)
- ii) If the selected student did not receive a scholarship, calculate the probability that they are enrolled in a post-graduate program. (6)

- b) Derive the Maximum Likelihood Estimator (MLE) for the parameter λ of a Poisson distribution. (8)
3. a) A research team at SASTRA University is analyzing student engagement using a 2D binary classification dataset. Highly engaged students are represented by points (2, 3) and (3, 4) (Positive Class, +1), while less engaged students are represented by points (1, 1) and (2, 1) (Negative Class, -1). Plot the points on a 2D graph and calculate the equation of the optimal separating hyperplane using a linear Support Vector Machine (SVM). Identify the support vectors that define the margin. (15)
- b) Explain the key differences between Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) in terms of their assumptions and the form of their decision boundaries. Derive the formula for the decision boundary in LDA and QDA for a two-class classification problem, assuming normal distributions with class-specific covariance matrices. (5)
4. a) Explain how convolution operations are performed in CNNs, the significance of filters (kernels), and how features are extracted. Also, describe how parameters like stride, padding, and filter size affect the output. (10)
- b) Describe how to identify whether a model is overfitting or underfitting during training. Outline strategies to mitigate both overfitting and underfitting in machine learning models. (5)
- c) Define ensemble learning and describe its primary goal. Explain two popular ensemble methods and how they contribute to model performance. (5)
5. a) Compute the Fisher's Linear Discriminant Analysis for the following 2D dataset: (15)
Samples for Class 1: {(4,1), (2, 4), (2,3) (3,6), (4,4)}
Samples for Class 2: {(9,10), (6, 8), (9,5) (8,7), (10,8)}

b) Explain the concept of Multiclass Logistic Regression using the SoftMax Function. How does it differ from binary logistic regression? (5)

6. a) Consider a POS-Tagger HMM that tags words with their corresponding Part-of-Speech (POS). It uses three hidden states: Noun (N), Verb (V), and Adverb (A). The following data helps the model make predictions:

(i) Transition Probabilities:

	N	V	A
N	0.2	0.5	0.3
V	0.3	0.4	0.3
A	0.4	0.4	0.2

(ii) Emission Probabilities:

	John	reads	very	fast
N	0.9	0.1	0.0	0.0
V	0.0	0.7	0.1	0.2
A	0.0	0.0	0.9	0.1

(iii) Initial Probabilities:

	N	V	A
N	0.5		
V	0.3		
A	0.2		

Illustrate the state-transition diagram based on the provided transition probabilities. And Utilize the Viterbi algorithm to find the most likely sequence of POS tags for the observation sequence "John reads very fast". (15)

b) Illustrate how the hidden state is updated in a GRU network. Provide the mathematical formulae and explain its components. (5)

PART – B

Answer the following

1 x 20 = 20 Marks

7. a) Consider a neural network configured for the task of image classification. The network consists of several neurons that employ the sigmoid activation function. Given the provided weights and biases for each neuron, perform the following tasks: (15)

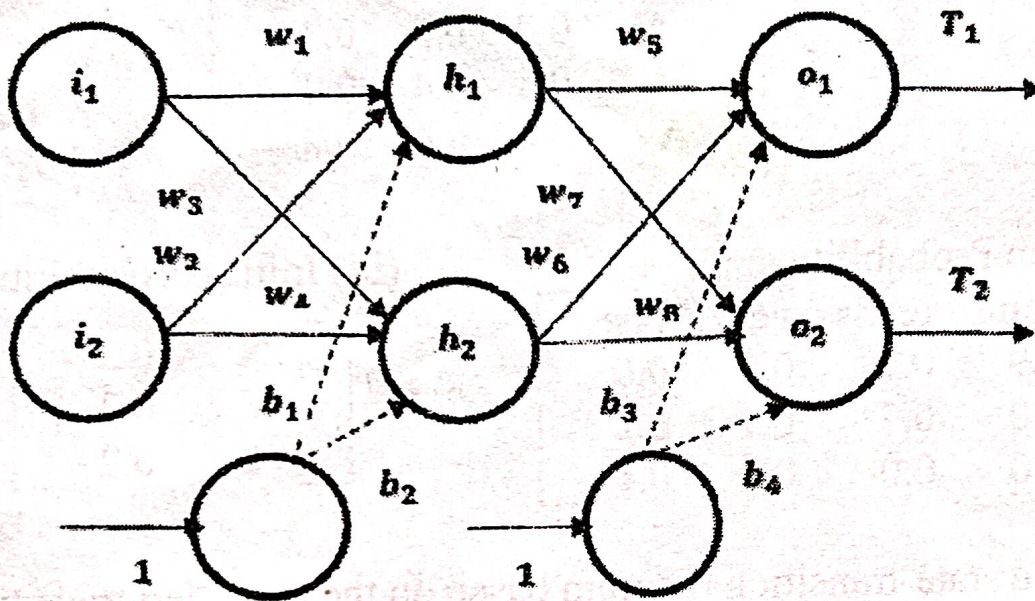
Input Features:
 $i_1 = 0.05$, $i_2 = 0.10$

Bias
 $b_1 = 0.35$, $b_2 = 0.35$, $b_3 = 0.60$,
 $b_4 = 0.60$

Weights:

$w_1 = 0.15$, $w_2 = 0.20$, $w_3 = 0.25$,
 $w_4 = 0.30$, $w_5 = 0.40$, $w_6 = 0.45$,
 $w_7 = 0.50$, $w_8 = 0.55$

Target Values:
 $T_1 = 0.01$, $T_2 = 0.99$



i) Perform a forward pass and a backward pass through the network, assuming a learning rate of 0.5.

ii) After updating the weights and biases from the backward pass, perform one iteration of the forward pass to verify whether the error is minimized?

b) A tech company is developing a predictive text application that suggests the next word based on previous ones. To handle long-term dependencies, they choose an LSTM network. How do LSTMs help overcome long-term dependency challenges in text prediction, and what mechanisms make them better suited than Vanilla RNNs for distant context? (5)

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